

# 4D Decomposition

Overlapping Rules → Disjoint Cells → TRF

## Input LRF (first-match order)

r1: TCP 10.0.0.x / 80–1023 → DENY

r2: IP ANY → ALLOW (r2 generalizes r1)

↓  
sweep-line 3D

## After 4D Decomposition (disjoint cells)

*(UDP and ICMP: one ALLOW cell each, inherited from r2)*

**TCP · src ≠ 10.0.0.x**

port 0–65,535

→ **ALLOW**

**TCP · 10.0.0.x**

port 0–79

→ **ALLOW**

**TCP · 10.0.0.x**

port 80–1023

→ **DENY**

**TCP · 10.0.0.x**

port 1024–65,535

→ **ALLOW**

↓  
directional merge

## TRF Output (TCP subtree, Ordering 4)

[protocol]=TCP → [dst\_ip]=ANY → [dst\_port] → [src\_ip]

port 0–79 → ALLOW

port 1024–65,535 → ALLOW

port 80–1023 → [src\_ip]: 10.0.0.x → DENY, other → ALLOW

*Every packet reaches exactly one leaf; the DENY region equals  $M(r1)$ , as first match requires.*